

LearnFlux AI

The Next-Generation Intelligent Adaptive Learning & Cognitive Telemetry Engine

Product: **LearnFlux AI** Status: **Production Ready Prototype** Live Application: <http://localhost:5173/>

Target: **Hackathon Grand Prize Pitch**

⚡ 1. The 60-Second Elevator Pitch

*"Right now, global education is broken because it treats millions of unique minds like identical assembly-line machines. Whether you are on Coursera, Khan Academy, or in a traditional classroom, everyone is fed the exact same static video or quiz. If you fail, the system just says 'Wrong: -1 point'—never telling you **why** your brain made that mistake.*

*Introducing **LearnFlux AI**—an intelligent adaptive learning engine that replaces static courses with an evolving **Prerequisite Knowledge Graph**. LearnFlux doesn't just score you; it extracts your **'Mistake DNA'**, classifying errors into conceptual misconceptions, calculation slips, careless mistakes, or application errors. Powered by a real-time **Mastery Engine**, an interactive **3D spatial WebGL canvas**, and a multimodal learning hub, LearnFlux identifies your exact bottleneck and prescribes the single **'Next-Best-Action'** to unlock mastery."*

🎤 2. The 3-5 Minute Stage Presentation Script

Act I: The Hook & The Problem (0:00 – 0:45)

*"Judges, have you ever spent 3 hours studying a topic, only to fail the exam because you lacked a single foundational prerequisite taught two years ago? 85% of students who drop out of STEM do so not because they aren't smart, but because of **hidden prerequisite knowledge gaps** that traditional courses never detect. Traditional EdTech platforms are passive, linear, and blind to how the human brain actually retains concepts."*

Act II: The Solution — LearnFlux AI (0:45 – 1:30)

"We built LearnFlux AI. When a learner logs on—whether a 10-year-old mastering fractions or a university senior preparing for placement exams—LearnFlux runs a rapid Diagnostic Knowledge Assessment. Behind the scenes, our graph algorithms map their responses directly against a prerequisite dependency tree, instantly generating an individualized Learning Path calibrated to their cognitive velocity."

Act III: The Breakthrough Innovations (1:30 – 2:45)

1. Mistake DNA™: Categorizes errors into Conceptual, Calculation, Careless, or Application errors instead of binary 0/1 scores.

2. Next-Best-Action Engine: Eliminates decision fatigue by pinpointing the single prerequisite bottleneck that unlocks the most downstream topics.

3. Multi-Modal Learning Hub: Serves 4 parallel formats for every concept: Theory Notes, Visual Mind Maps, Active Flashcards, and Adaptive Quizzes.

Act IV: Market & Closing Call-to-Action (2:45 – 4:00)

"The Smart EdTech market is projected to reach \$400B+ by 2030. Our B2C freemium model (\$12/mo) pairs with institutional B2B licensing for colleges and school districts. LearnFlux turns education from a passive one-way broadcast into an intelligent, adaptive dialog."

3. Detailed Concept & Algorithmic Mechanics

The 4 Cognitive Mistake DNA™ Axes

- **Conceptual** (E_{concept}): Missing baseline theorem or axiomatic definition. *Action: Direct to theory notes & mind maps.*
- **Calculation** (E_{calc}): Mechanical arithmetic or syntax error. *Action: Flashcard recall drills.*
- **Careless** (E_{careless}): Misread prompt under time pressure. *Action: Pacing drills.*
- **Application** (E_{apply}): Correct formula applied to wrong context. *Action: Diagnostic case quizzes.*

Prerequisite Knowledge Graph & Bottlenecks

Concepts are modeled as a Directed Acyclic Graph (DAG). If a student struggles in *Dynamic Programming*, LearnFlux traverses backward to test *Recursion*, *Binary Search*, and *Big-O*. It isolates the highest-leverage node—resolving one bottleneck unlocks multiple downstream milestones.

Accuracy-Weighted Mastery Score Formula

Rather than a crude percentage, LearnFlux computes student mastery via a dynamic difficulty-weighted equation:

$$\text{Mastery} = \min(100, \max(0, \text{Accuracy} \times 100 \times (0.7 + 0.3 \times (\text{AvgDifficulty} / 5))))$$

Where Accuracy is evaluated over the last 10 attempts, and AvgDifficulty scales from 1 to 5. Scores $\geq 80\%$ mark the node **Mastered**; 50–79% marks it **Developing**; $< 50\%$ flags an active **Knowledge Gap**.

4. Complete Technical Stack Breakdown

Frontend & UI Architecture

Technology	Category	Version	Role & Implementation in LearnFlux AI
React	UI Framework	^19.1.1	Component lifecycle, concurrent rendering, dynamic state hydration, hooks, and responsive layout management.
Vite	Build & Dev Tool	^7.1.7	Sub-millisecond Hot Module Replacement (HMR) and optimized ESM asset bundling.
Three.js	3D WebGL Core	^0.180.0	Low-level WebGL graphics engine, 3D matrix math, geometry, camera transforms, and lighting pipelines.
@react-three/fiber	React 3D Bridge	^9.3.0	Declarative Three.js scene graph rendering directly inside React's virtual DOM tree.
@react-three/drei	3D Utilities	^10.7.6	Powers OrbitControls, Float dynamics, 3D particle systems (Sparkles), and mesh management.
Lucide React	Iconography	^0.468.0	Pixel-perfect iconography for telemetry badges, status indicators, and course catalog streams.
Custom CSS3	UI Design System	CSS3 / BEM	Neon glassmorphism, responsive grid/flexbox layouts, CSS variables, and real-time cursor light aura (CursorGlow).
Web Speech API	Audio & Voice	Native API	Browser-native SpeechSynthesis for real-time text-to-speech voice narration in Flux Bot and AI Tutor.
Cursor Shaders	Spatial Shaders	Custom Math	Normalized coordinate interpolation (lerp) controlling 3D mouse spotlights and animated 3D pupil gaze.
SVG / HTML5 Canvas	Data Visualization	Native SVG	Smooth bezier curve learning velocity charts and 6-axis cognitive radar polygons in Progress Tracker.

Backend, Database & Algorithmic Stack

Technology	Category	Version / Spec	Role & Implementation in LearnFlux AI
Supabase (PostgreSQL)	Primary Database	PostgreSQL 15+	Relational Knowledge Graph modeling: profiles, subjects, topics, subtopics, study_materials, questions, attempts.
PostgreSQL pgcrypto	Cryptographic Engine	Core Extension	Generates cryptographically secure UUIDv4 primary keys across all relational tables.
Row-Level Security (RLS)	Data Security	Zero-Trust Policies	Enforces multi-tenant isolation: students can only access and modify their own attempts and progress via auth.uid().
Deno / TypeScript	Serverless Compute	Supabase Edge	Executes serverless Edge Functions (record-attempt, get-full-profile, get-progress-over-time) with <50ms cold starts.
Mastery Engine	Algorithmic Core	masteryEngine.ts	Calculates dynamic accuracy-weighted scores and manages status transitions (Locked → Weak → Developing → Mastered).
Mistake DNA™ Classifier	Cognitive Profiler	Custom Engine	Parses quiz distractor error taxonomies in real time to update student cognitive breakdown.
Universal Hybrid Client	Data Access Layer	backendApi.js	Dual-mode connector: operates against live Supabase endpoints with instant local fallback for zero-fail hackathon demos.
Supabase & Firebase Auth	Identity Provider	JWT / Token	Manages user onboarding, authentication sessions, grade-level tagging (Grade 1–12), and target learning goals.
Multi-LLM Orchestrator	AI Reasoning Layer	Swappable API	Connects to fine-tuned LearnFlux models, GPT-4o, Claude 3.5 Sonnet, and Gemini Pro with 4 pedagogical personas.
Prerequisite DAG Router	Graph Engine	Directed Graph	Traverses prerequisite node relationships to compute the critical path bottleneck and assign Next-Best-Action focus.

5. The 6-Step Hackathon Live Demo Flow

- Step 1: The Visual Hook** — Showcase the interactive 3D crystal core rotating at 60fps on the landing page (<http://localhost:5173/>).
- Step 2: Diagnostic Assessment** — Launch a subject diagnostic test; intentionally answer with a distractor to demonstrate real-time Mistake DNA error classification.
- Step 3: Adaptive Learning Path** — Show how the curriculum dynamically rearranged itself to isolate the detected bottleneck (*Time Complexity & Big-O*).
- Step 4: Interactive 3D Sphere & Multi-Modal Hub** — Navigate to the Dashboard; hover mouse to show the 3D Sphere tracking cursor vectors, spotlighting, and pupil gazing. Scroll down and toggle between Theory Notes, Visual Mind Maps, and Interactive Flashcards.
- Step 5: Adaptive Knowledge Quiz** — Complete a 3-question sequence showing streak multipliers and dynamic difficulty scaling from Foundation to Intermediate.
- Step 6: Voice Flux Bot & AI Tutor** — Click Flux Bot on the dashboard to hear vocal speech greeting, then ask the AI Tutor a technical query with multi-persona switching and one-click Markdown chat export.

? 6. Anticipated Judge Questions & Winning Defenses

Q: "How is this different from Khan Academy or Duolingo?"

A: Khan Academy and Duolingo are linear tracks with binary right/wrong scoring. LearnFlux diagnoses *why* you failed using Mistake DNA (conceptual vs careless vs calculation) and provides multi-modal learning modalities (mind maps, notes, flashcards) for every single node.

Q: "Can LearnFlux support custom university syllabi or notes?"

A: Yes! LearnFlux features a '+ Add Subject' engine where students or educators can upload lecture notes and syllabi, which automatically synthesize into personalized diagnostics and prerequisite learning paths.

Q: "How is student data protected?"

A: Full enterprise-grade PostgreSQL Row-Level Security (RLS) is implemented on Supabase. Every telemetry point, attempt, and progress record is bound to `auth.uid()`, preventing any unauthorized cross-tenant data leaks.